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Deep learning for oncologic treatment outcomes and endpoints evaluation from CT scans in liver cancer



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Accurate treatment response assessment using serial CT scans is essential in oncological clinical trials. However, oncologists' assessment following the Response Evaluation Criteria in Solid Tumors (RECIST) guideline is subjective, time-consuming, and sometimes fallible. Advanced liver cancer often presents multifocal hepatic lesions on CT imaging, making accurate characterization more challenging than with other malignancies. In this work, we developed a tumor volume guided comprehensive objective response evaluation based on deep learning (RECORD) for liver cancer. RECORD performs liver tumor segmentation, followed by sum of the volume (SOV)-based treatment response classification and new lesion assessment. Then, it can provide treatment evaluations of response, stability, and progression, and calculates progression-free survival (PFS) and response time. The RECORD pipeline was developed with both CNN and ViT backbones. Its performance was evaluated in three longitudinal cohorts involving 60 multi-national centers, 206 patients, 891 CT scans, using internal five-fold cross-validation and external validations. RECORD with the most effective backbone achieved an average AUC-response of 0.981, AUC-stable of 0.929, and AUC-progression of 0.969 for SOV-based disease status classification, F1-score of 0.887 for new lesion identification, and accuracy of 0.889 for final treatment outcome assessments across all cohorts. RECORD's PFS and response time predictions strongly correlated with clinician's assessments ($P < 0.001$). Moreover, RECORD can better stratify high-risk versus low-risk patients for overall survival compared to the human-assessed RECIST results. In conclusion, RECORD demonstrates efficiency and objectivity in analyzing liver lesions for treatment response evaluation. Further research should extend the pipeline to other metastatic organ sites.

Objective response rate and progression-free survival (PFS) are the commonly used endpoints in phase II/III clinical trials of anticancer drugs^{1,2}, which rely heavily on accurate assessment of treatment outcomes. The Response Evaluation Criteria in Solid Tumors (RECIST) guideline (version 1.1) is the currently established, standardized approach for assessing treatment response based on changes in tumor burden³. RECIST defines four

principal tumor response outcomes: complete response (CR), partial response (PR), stable disease (SD), and progressive disease (PD), according to the changes in tumor diameter between baseline and follow-up scans. In clinical practice, radiologists must perform several steps to evaluate treatment response: identify lesions on serial imaging, select target lesions, measure changes in tumor burden on serial scans, and make optimal

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therapeutic decisions. The procedure is extremely labor-intensive and time-consuming. Furthermore, the subjectivity of image-reading procedures can lead to discrepancies between radiologists. Studies have reported discordance rates ranging from 23% to 46% and 42% between two readers^{4,5}. Inaccurate assessments can lead to wrong medical decisions, thereby compromising treatment efficacy and shortening survival time.

Currently, artificial intelligence (AI) models have been widely applied to automatically quantify the tumor burden from CT scans, such as measuring the diameter^{6,7} and segmenting the 3D mask^{8–10} of a specific lesion, in seconds. Consequently, both linear and volumetric measurements can be efficiently obtained, where it was previously arduous for oncologists to manually draw tumor contours on each slice in the clinical practice. Besides, several researchers have investigated the feasibility of volumetric measurements to quantify tumor burden as extensions to unidimensional RECIST approaches in a range of cancer types^{11–15}.

Advanced liver cancer often presents with multiple nodules within the liver on CT imaging, which can be more difficult to identify lesions accurately compared to other cancer types. Together with the complex anatomy of the abdominal cavity, low image contrast between organs, unclear boundaries of primary liver tumors, and the heterogeneity of liver cancer subtypes (nodular, massive, and diffuse) that exhibit different radiological characteristics^{16,17}, accurate treatment outcome measurements using AI models can be especially challenging. Though there existed a series of automatic AI methods for tumor segmentation and detection, none of them provide treatment outcomes and estimate clinical endpoints by systematical analyses of the downstream tumor changes and new lesions. In addition, adhesion between lesions in advanced liver cancer can lead to variations in diameter calculations. Specifically, the sum of the diameters of the two individual lesions, when calculated separately, will differ from the diameter calculated when the two interconnected lesions are considered as a single lesion. Moreover, a growing field of research has demonstrated that volumetric measurements are more effective predictor for therapeutic responses than the diameter-based RECIST approach^{18–20}. Therefore, we considered to employ volumetric measurements for evaluating treatment response in patients with advanced liver cancer.

However, a direct application of current (SOTA) segmentation models may lead to measurement errors compounded in treatment outcome assessment scenarios, as comparisons of tumor burden between baseline and follow-up are necessary. Current segmentation models are prone to making inconsistent predictions for the same lesion in serial imaging studies^{21,22}. For liver cancer, suboptimal contrast enhancement of the imaging or ambiguous lesion boundaries can increase the unpredictability of the automatic tumor segmentation results predicted by SOTA segmentation models within a single scan^{23,24}. Therefore, employing the standalone segmentation model to separately delineate the baseline and follow-up scan could result in the compounded error from each individual segmentation result²⁵. The compounded inaccuracies may then engender an actual similar-trended lesion being erroneously and inconsistently predicted. That is, its actual size maintains stable, while the model mis-estimates which in CT series, thereby causing overestimation or underestimation of the tumor burden and incorrect objective response evaluations. Consequently, instead of predicting each CT scan independently like current segmentation models, a deep learning model for treatment outcome assessment should take into account the correlation-ship between baseline and follow-up scans to capture the change patterns of lesions across serial CT scans.

In this study, we developed an automatic treatment response assessment model for liver cancer with two key features: firstly, instead of using the RECIST approach of tracking the diameters of two lesions, we employed the sum of liver tumor volume (SOV) to evaluate treatment outcomes; secondly, in the model design, we incorporated the correlation between longitudinal CT scans into the traditional segmentation model. The developed deep learning model is a two-staged pipeline that mimics the radiologist's evaluation process in the clinical practices. In the first stage, the model can predict tumor masks using any state-of-the-art segmentation model. We used nnU-Net²⁶ and Swin-Unetr²⁷ as typical representatives. Then, a multi-

task learning architecture with several objective-response-specific methods is adopted to optimize potential inconsistent tumor predictions of SOTA segmentation models, thereby improving the tumor assessment accuracy. The model can assess tumor burden change between the baseline and follow-up scan by adding up the volume of all liver tumors, and identify new lesion appearances on the follow-up scan, then integrate the two to provide overall treatment response assessment outcomes. We validated the model performances at both image-pair level (i.e., treatment outcome evaluations on a baseline-follow-up CT pair) and the downstream patient level (i.e., endpoint measures of response time and PFS). To verify the effectiveness of the clinician-inspired architecture in our model, we compared to conventional state-of-the-art (SOTA) segmentation models on the evaluation accuracy. We further analyzed the correlations between the model-predicted PFS or response time and the overall survival, to back the clinical relevance of our developed RECORD pipeline.

Results

Dataset description

This retrospective study consisted of three longitudinal cohorts at 8 sites from 60 centers. Cohorts A and B represented clinical trials of tislelizumab. Cohort A consisted of patients with unresectable hepatocellular carcinoma enrolled from April 9, 2018 to July 6, 2022. Cohort B consisted of patients with metastatic liver cancer from urothelial bladder enrolled from June 16, 2017 to March 11, 2021. Cohort C included patients with hepatocellular carcinoma treated with single immunotherapy or two different immunotherapy drugs from Xinhua Hospital between July 18, 2017 and November 25, 2022. Sample selection process and inclusion and exclusion criteria are presented in Fig. 1.

A total of 206 immunotherapy-treated patients with liver cancer, 891 venous-phase CT scans were included. The model training (and five-fold cross validation) consisted of 111 patients from 33 centers. Three heterogeneous datasets consisted of 39 patients from 15 centers, 22 patients from 11 centers, and 34 patients from a single center were assigned external validations. Cohort characteristics is described in Table 1. Distribution of patient number and serial scan number in each cohort are presented in Supplementary Fig. 1.

Study design and model architecture

The treatment response outcome at each follow-up compared to baseline was assessed then calculated the downstream clinical endpoints evaluations (Fig. 2A). Due to the rarity of CR outcomes, CR and PR were combined as one type of response (hereafter referred to as PR)²⁸, i.e., three types of objective response outcomes (PR, SD, PD) were considered. The model was trained on Cohort A1 using five-fold cross-validation, then ensemble and assessed on three heterogeneous cohorts, i.e., Cohort A2, B, and C (Fig. 2B).

Tumor volume guided comprehensive objective response evaluation based on deep learning (RECORD) is an AI pipeline that modeled the radiologist's evaluation procedure in clinical practice (Fig. 2C). RECORD adopted a two-stage architecture that performs tumor segmentation (stage one, Fig. 3A) followed by sum of the volume (SOV)-based treatment response classification (stage two, Fig. 3B, upper branch) and new lesions assessment (stage two, Fig. 3B, lower branch). Then, RECORD can provide the overall treatment response outcome of a baseline-follow-up scan of a particular patient being either PR, SD, or PD according to match-up rules (Fig. 3B, the gray table).

In the initial tumor segmentation step, two SOTA models from major segmentation model backbones, i.e., convolutional neural networks (CNN)-based nnU-Net and Vision Transformer (ViT)-based Swin-Unetr, were employed to coarsely identify lesions on the venous-phase CT scan. Then, deformable registration was applied to align the baseline and follow-up images, compensating for potential physical shifts at different time points. The alignment allows the overlay of the follow-up image on the baseline image to directly reflect the tumor change. The difference map was specially designed to reflect and represent the difference between registered predicted baseline-follow-up masks. It was originally generated by subtracting the

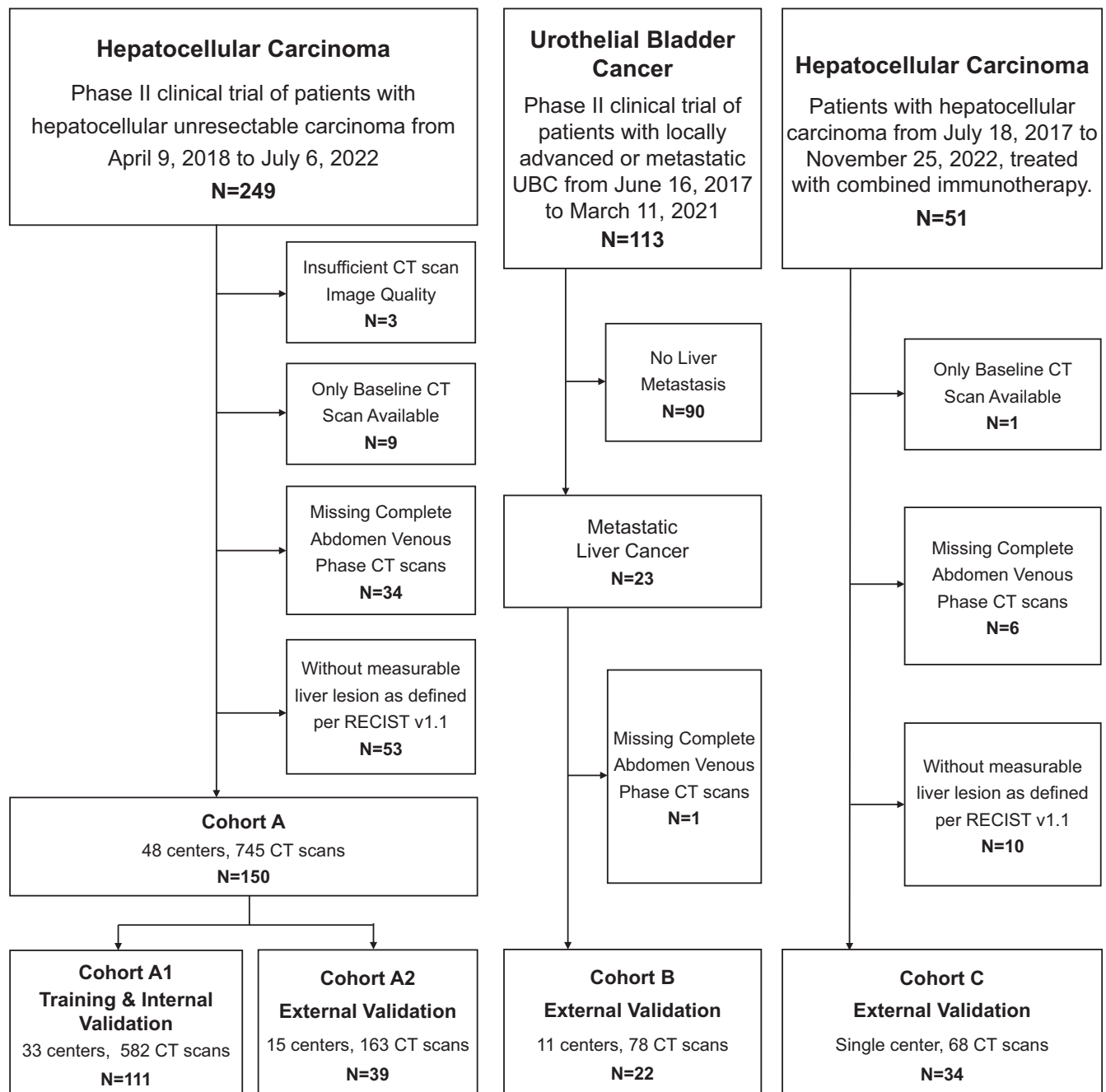


Fig. 1 | Flowchart of sample selection.

baseline mask from the follow-up mask, and adjusted with weak supervision from serial scans to incorporate longitudinal correlations and anatomical information (Fig. 3C). With the guidance of the difference map, RECORD can perform the SOV-based tumor burden change analysis and new lesion identification.

The SOV-based treatment response classification was built on a multi-task learning framework that considered both the accuracy of lesion segmentation and progression classification. In this sub-model, the SOV changes between baseline and follow-up tumor were transformed into a three-class probability of treatment outcome by a mapping function. The mapping function utilized a label encoding strategy tailored for evaluating response outcomes through ordinal regression, i.e., as the response transitions to progression, the status progressively worsens. The new lesion assessment used a graph-matching strategy based on Euclidean distance to associate the same lesion at different time points. By combining the

evaluation results from the SOV changes and the new lesion assessment (Supplementary Table 1), the final treatment outcome can be determined.

RECORD performances on the evaluation of treatment outcomes

The RECORD pipeline can accurately measure the treatment outcome, with the CNN backbone performing slightly better than the ViT backbone for combined results from Cohort A1, A2, B, and C. Specifically, for SOV-based progression status classification, the CNN-based RECORD showed the AUROC of 0.981 (95% CI: 0.971, 0.991) in response, 0.929 (95% CI: 0.908, 0.949) in stability, and 0.969 (95% CI: 0.957, 0.981) in progression. The ViT-based RECORD had the AUROC of 0.974 (95% CI: 0.960, 0.988) in response, 0.891 (95% CI: 0.864, 0.918) in stability, and 0.943 (95% CI: 0.925, 0.961) in progression. For new lesion assessment, Cohort C was excluded due to unclear definition of unequivocal new lesions in patients with terminal stage and extensive liver damage (>50%). The deep learning model

Table 1 | Cohorts characteristics

Characteristics	Internal Cohort A1	External Cohort A2	Cohort B	Cohort C	p-value	
					Univariate	Multivariate
Center	33	15	11	1		
Patient	111	39	22	34		
CT Scan	582	163	78	68		
Median serial scan number	4	3	2	2		
Median follow-up time (day)	510	467	312	NA ^a		
Median treatment time (day)	140	95	81	NA ^b		
Treatment response label (%)					0.334	0.124
Response	106 (23)	25 (20)	8 (14)	4 (12)		
Stable	194 (41)	60 (48)	25 (45)	19 (56)		
Progression	171 (36)	39 (32)	23 (41)	11 (32)		
Sex (%)					0.191	0.067
Male	93 (84)	36 (92)	16 (73)	32 (94)		
Female	18 (16)	3 (8)	6 (27)	2 (6)		
Age (year) (%)					0.248	0.104
<50	12 (11)	6 (15)	1 (4)	9 (26)		
50–59	32 (29)	10 (26)	9 (42)	9 (26)		
60–69	36 (32)	14 (36)	11 (50)	10 (30)		
70–79	22 (20)	7 (18)	1 (4)	5 (15)		
>80	9 (8)	2 (5)	0 (0)	1 (3)		
Tumor statistics, mean ± std						
Overall tumor volume (cm ³)	159.79 ± 350.65	247.27 ± 509.65	97.31 ± 326.62	189.90 ± 374.92	0.064	0.399
Lesion number per patient	4.86 ± 6.03	5.42 ± 6.64	3.79 ± 5.22	3.65 ± 5.10	0.095	0.094
Single lesion volume (cm ³)	32.73 ± 20.32	45.49 ± 28.42	25.47 ± 19.58	51.77 ± 26.57	0.100	0.596

Univariate analysis was conducted using Chi-square test for discrete variables, and the ANOVA for continuous variables. Multivariate analysis used the ANOVA method.

Tumor statistics refer to the liver lesions.

NA not available, std standard deviation.

^{a,b}Treatment and follow-up times were unavailable for Cohort C.

achieved the best precision of 0.901 (95% CI: 0.875, 0.921), recall of 0.874 (95% CI: 0.846, 0.897), F1-score of 0.887 (95% CI: 0.861, 0.909), and accuracy of 0.954 (95% CI: 0.935, 0.968) on CNN-based RECORD (Fig. 4A, the middle panel). For overall treatment response outcomes which assess both SOV-based tumor burden change and presence of new lesions, the best accuracies were 0.873 (95% CI: 0.839, 0.900) on the internal validation dataset, and 0.927 (95% CI: 0.868, 0.961), 0.982 (95% CI: 0.906, 0.997), 0.882 (95% CI: 0.734, 0.953) on three external validation datasets—cohort A2, B, and C, respectively.

To visually illustrate the prediction performance, the segmentations and treatment outcomes of patients with serial scans predicted by the RECORD model from both internal and external validations were randomly selected and presented (Fig. 5 and Supplementary Fig. 2). RECORD can generate reliable tumor segmentation masks as well as three-class progression classification results.

Besides, the efficiency of RECORD was evaluated (Supplementary Fig. 3). We have calculated the time consumption of each step in the prediction process. The average total processing time for RECORD was 16.18 [SD 2.71] seconds for evaluating the treatment outcome.

Validity of the clinician-inspired RECORD architecture

To verify the validity of the specially-designed clinician-inspired architecture in the RECORD pipeline, treatment assessment accuracies were compared between SOTA segmentation models and our proposed pipeline (i.e., nnU-Net vs CNN-based RECORD, Swin-Unetr vs ViT-based RECORD). The SOTA segmentation model refers to a standalone model that segments each CT scan independently to provide only tumor masks

(Fig. 2C, original segmentation step; Fig. 3A). The tumor burden change and new lesion appearance, two components for the overall treatment outcome evaluation, were calculated based on these masks.

For the overall treatment response accuracy, the CNN-based RECORD had an improvement of 3.50% compared to nnU-Net, the ViT-based RECORD had an improvement of 3.07% compared to Swin-Unetr. Compared to the segmentation models, the RECORD pipeline incorporates deformable registration and a difference map into a multi-task learning architecture in addition to the original segmentation step (Figs. 2C and 3B). The integration captures the correlation among serial scans that is neglected in standalone SOTA segmentation models, contributing to a significant improvement in treatment assessment accuracies (Tables 2 and 3, all $P < 0.01$). Detailed AUC curves and confusion matrices are shown in Supplementary Figs. 4–7 for RECORD, and Supplementary Figs. 8–11 for segmentation models.

Clinical endpoints evaluations

To further evaluate the clinical utility of RECORD, the performance of both CNN-based and ViT-based models on the assessment of clinical endpoints (i.e., PFS and time to response) was investigated. Clinician-derived, RECORD-predicted, and human-assessed RECIST v1.1 clinical endpoints were compared. Clinician-derived clinical endpoints were determined by applying the mapping function to the annotated tumor volume on each follow-up, then calculated the PFS and response time. The RECIST v1.1 results were determined by a separate independent review committee that differed from the underlying clinical trial by strictly following the RECIST guideline. Correlations of the two endpoints among the human-assessed

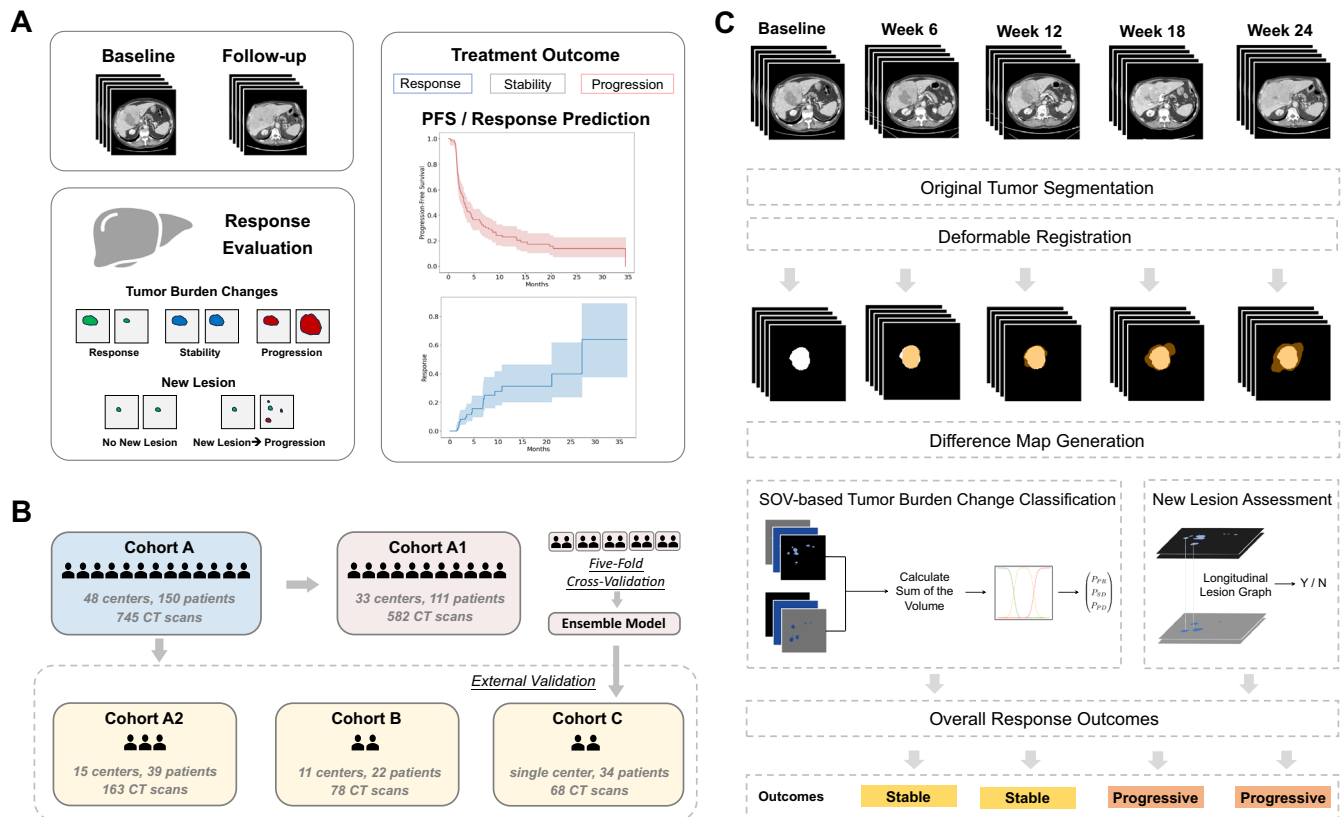


Fig. 2 | Overall study design and workflow of the proposed automatic treatment response assessment model. A Schematic overview of the developed deep learning model for automatic treatment assessments and clinical endpoints evaluation from serial CT scans. **B** Training and validation strategies. **C** Procedures of the proposed

RECORD model to provide overall response outcomes of a patient with serial CT scans. CR complete response, PR partial response, SD stable disease, PD progressive disease.

RECIST v1.1, clinician-derived, and model-predicted were visualized and calculated (Fig. 6 and Table 4).

The CNN-based RECORD model had reached c-indexes of 0.958 (95% CI: 0.929, 0.987) and 0.954 (95% CI: 0.908, 1.000) with clinician's annotation-derived PFS and response time, respectively. Spearman coefficients also showed strong positive agreements. The ViT-based RECORD model exhibited similar results. These results suggested that the model can yield statistically significant correlations with the clinician's measures (all $P < 0.001$). Besides, volumetric-based methods also exhibited appreciable agreements with the conventional RECIST v1.1 criteria.

Correlations between clinical endpoints and survival

Since the overall survival (OS) served as the final goal for cancer patients, correlations between model-predicted PFS and response time and the overall survival were further analyzed to back the clinical relevance of the developed RECORD treatment outcome assessment pipeline. Landmarking analyses revealed that our model-predicted disease progression at 3 and 6 months were both predictive of OS (Fig. 7A, B, all $P < 0.001$). Similar to PFS, results indicated that time to response can serve as a predictor of OS (Fig. 7C, $P < 0.001$). Compared to risk stratification by segmentation model-derived results (Supplementary Fig. 12), RECORD can achieve larger hazard ratios (HR) and area between survival curves (ABS). Besides, the results demonstrated that the RECORD assessment, which is based on the sum of lesion volumes, had superior risk stratification capabilities for overall survival compared to the manual RECIST v1.1 assessment that only considers the two largest lesions. Specifically, when using progression at 3 months assessed by different measures to make risk stratification, SOV-based method had higher hazard ratios and ABS (Fig. 7 and left panel of Supplementary Fig. 13, clinician-derived, HR, 2.332, 95% CI: 1.539–3.534; ABS, 9.107; CNN-based RECORD, HR 1.991, 95% CI: 1.327–2.989; ABS, 7.641;

ViT-based RECORD, HR 2.204, 95% CI 1.470–3.305; ABS, 8.545) than the RECIST criteria (right panel of Supplementary Fig. 13, HR, 1.922, 95% CI: 1.288–2.867; ABS, 6.939).

Discussion

To address the subjectivity and improve the efficiency of treatment outcome evaluations in clinical practice, we presented RECORD, a deep learning pipeline that facilitates the objective evaluation of treatment response. RECORD presents a scientific approach to convert the radiologist's evaluation process into an AI pipeline. It significantly outperformed conventional SOTA segmentation models ($P < 0.001$) in the accuracy of response outcome evaluations, and yielded statistically strong agreements with clinicians on PFS and response time assessments ($P < 0.001$). Moreover, the RECORD assessment, which based on the sum of lesion volumes, exhibited superior risk stratification capabilities for overall survival compared to the manual RECIST v1.1 assessment that only considers the two largest lesions. Compared to the arduous procedure for clinicians to delineate tumor contours on 3D CT scans and then make therapeutic decisions, RECORD can complete the entire process within a minute. These findings provided a starting point for using AI to evaluate tumor response outcomes.

Research on AI-based objective response assessment is important but scarce. Most relevant studies^{29–31} have focused on measuring the diameter or tumor volume from a single CT scan, without providing longitudinal analyses. Whereas, some longitudinal lesion identification works^{32,33} cannot provide a global evaluation of the overall tumor burden. Some recent studies^{25,34,35} have developed segmentation models for response assessment in different cancer types, such as lung, liver, and breast. However, these models are all simplifications of RECIST by considering either SOV change or new lesion appearances. A study³⁶ utilized deep learning to automate the measurement and tracking of lesion diameters in the RECIST progression

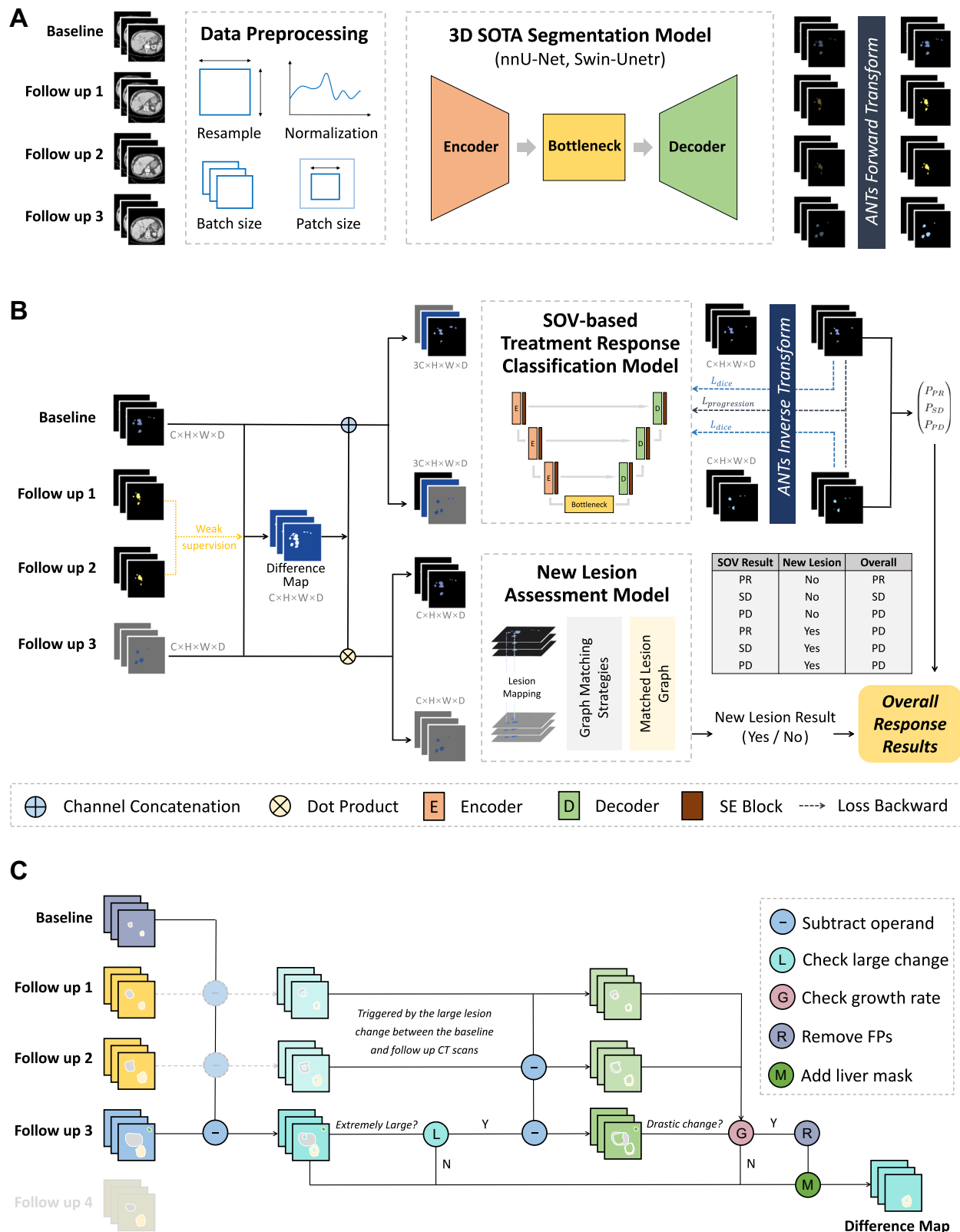


Fig. 3 | Overview of the RECORD architecture. **A** The tumor segmentation step. **B** The SOV-based response evaluation model and new lesion assessment model. Taken two parts together, RECORD can provide the final overall response outcome.

C Difference map generation process. ANTs advanced normalization tools, SOV sum of the volume, C channel, H height, D depth, W width, PR (complete) or partial response, SD stable disease, PD progressive disease.

assessment process. Whereas, this work had a small sample size, including only 54 patients with serial CT scans, and did not account for the identification of new lesions. Moreover, it struggles with complex or ambiguous lesion scenarios. Besides, all these models developed in current studies do

not consider the correlation of serial scans. The lack of consideration of correlation in serial scans of these models may lead to false-positive or false-negative predictions among the same longitudinal tumor and make the model inaccurate for the evaluation of patients in advanced or terminal

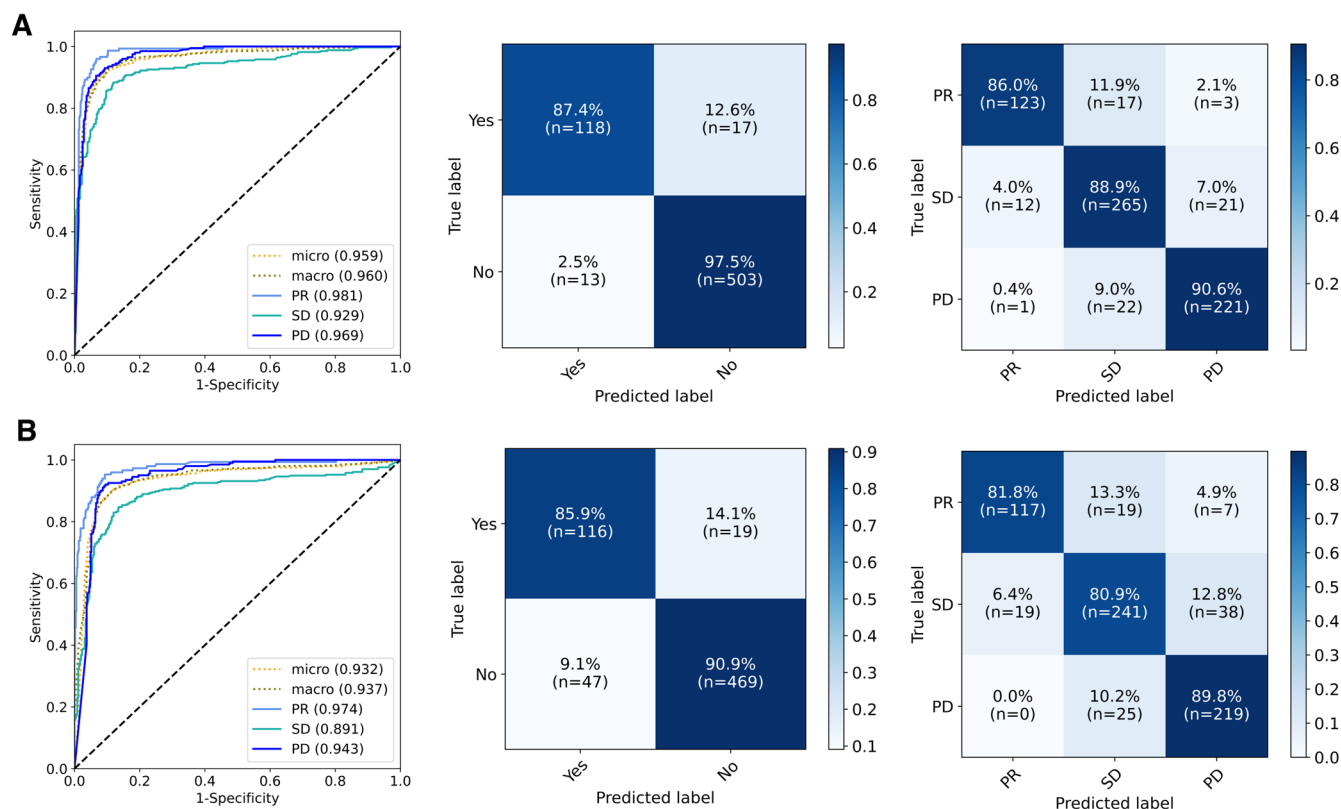


Fig. 4 | Performances of the RECORD pipeline. Performances of CNN-based (A) and ViT-based RECORD (B) on all validation sets. Five-fold cross-validation results in cohort A1, and results in Cohort A2, B, and C are merged. Left, AUROC of the SOV-based tumor progression status classification. Middle, Confusion matrices of

the new lesion assessments. Right, Confusion matrices of the overall treatment response outcome. PR (complete) or partial response, SD stable disease, PD progressive disease.

stages. Our proposed RECORD pipeline leveraged prior medical knowledge of serial CT scan correlation and anatomical information and employed a specially designed ‘volume to three-class probability’ mapping function, enabling it to provide accurate and objective treatment response assessment results.

RECORD represents the radiologist’s response assessment procedure in clinical practice. Specifically, the first step of the original tumor segmentation corresponds to the radiologist’s first step for identifying all lesions on the baseline and follow-up scans. Next, the deformable registration of predicted lesions reflects the process of the human eye aligning multiple serial images. Human eyes can repeatedly examine multiple serial images to achieve alignment, while algorithm-based models require registration to align the scans. We adopted inverse-consistent deformable transforms, thereby the model can directly learn the lesion growth trend as well as figure out the appearance of new lesions based on the overlay of registered images and masks. Then, the difference map generated from the registered segmentation results describes the process by which oncologists compare lesions from two time points to identify differences. Apart from the discrete three-class output of treatment outcomes, RECORD can also provide a probability for each class based on the SOV changes between baseline and follow-up scans. The continuous probability provides clinicians with a more informative quantification of treatment results than the discrete one, especially when progression or response is borderline.

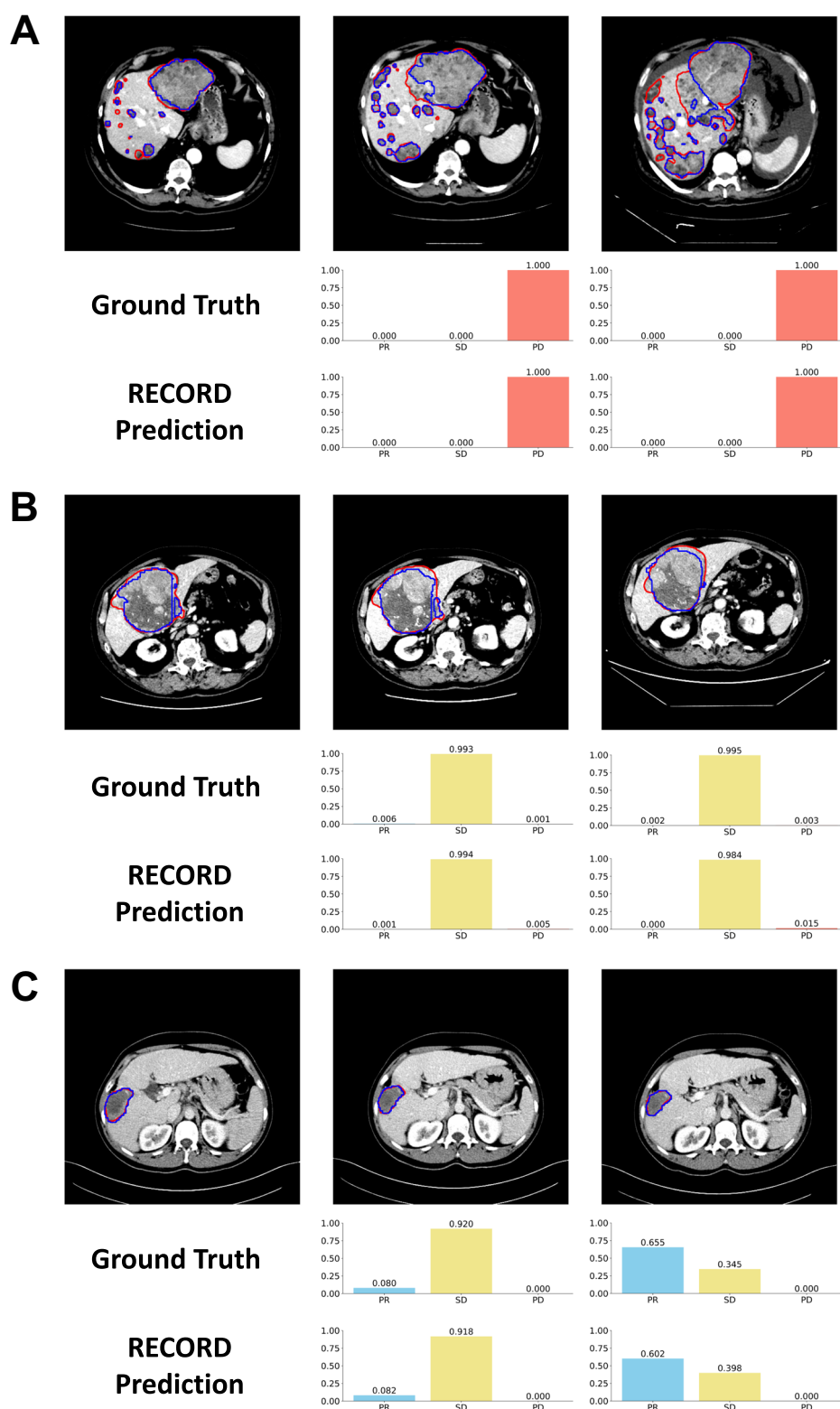
Our proposed RECORD pipeline not only achieves high accuracy in the baseline-follow-up level assessments, but also demonstrates high consistency with clinicians in patient-level PFS and response time evaluations. In clinical trials for liver cancer drugs, PFS and objective response rate are commonly used outcome measures³⁷, while accurate assessment of the response and progression status forms the basis for these measures. Our developed model can assist in clinical outcomes

judgments, reduce the subjectivity of assessments among doctors, and promote consistency treatment efficacy evaluations across various medical centers.

Overall survival is considered the most important outcome measure for assessing the efficacy of a drug, but it may not always be feasible to measure in all clinical trials. Therefore, surrogate endpoints like PFS or response time are often used to estimate the drug’s impact on overall survival. However, the validity of these surrogate endpoints requires further investigation^{37,38}. The RECORD provides a starting point for AI predictions being surrogates for the OS. Our findings show a significant correlation between the model’s assessment of tumor response/progression and overall survival in patients with unresectable HCC receiving immunotherapy as a second or third-line treatment. Nevertheless, our analysis was conducted at single-trial level, trial-level associations should be further explored to enhance the effectiveness.

This study has some limitations. Firstly, we excluded metastatic lesions in other organs. Due to the dispersed locations of metastatic lesions, we did not have sufficient samples to train separate segmentation models for each organ. If large training samples were available, we could construct a comprehensive multi-organ model to obtain more accurate and complete measurements. Secondly, a larger prospective cohort is needed to better validate the performance of RECORD in real-world settings. Thirdly, although our findings suggest that using volume as a measure of disease progression may serve as a better surrogate for survival compared to diameter measurements based on RECIST, the limited sample size may introduce bias into the conclusions. Currently, RECIST remains the recognized surrogate endpoint for survival, which has been validated in multiple large studies. The effectiveness of volume-based response endpoints as a surrogate for survival outcomes still needs to be validated in larger cohorts. Fourthly, the volumetric measurement of treatment outcome for the deep learning model developed in this study is

Fig. 5 | Cases for patients with serial CT scans. Red, ground truth. Blue, model predictions. Three cases of PD, SD, and PR are shown in the figure. Predictions of three-class probabilities are shown in the bar charts. Case (A) is from Cohort A1, (B) is from Cohort A2, (C) is from Cohort B. PR (complete) or partial response, SD stable disease, PD progressive disease.



extrapolated from the RECIST criteria. However, the modified RECIST (mRECIST)³⁹, which emphasizes viable tumor tissue, have been widely adopted in clinical guidelines for HCC. The future research should consider expanding the current model to include the enhanced portion of lesions, thereby enhancing its applicability for assessing the treatment response for liver cancer.

In conclusion, our proposed deep learning pipeline is capable of automatically reading 3D CT scans and then performing a

comprehensive treatment response assessment with high accuracy, objectivity, and efficiency. However, the current RECORD pipeline is limited to only analyzing liver lesions. Further research needed to expand the applicability of RECORD to assess metastatic sites. With continued advancement, RECORD has the potential to assist oncologists in monitoring patient disease status and to help standardize treatment response assessment in clinical trials and routine clinical practice.

Table 2 | Comparisons of treatment response outcomes on validation sets (nnU-Net vs CNN-based RECORD)

Metrics	Cohort A1	Cohort A2	Cohort B	Cohort C
<i>nnU-Net</i>				
SOV-based Tumor Burden Change (benchmark)				
AUC-PR	0.961 (0.938, 0.985)	0.978 (0.956, 1.000)	1.000 (1.000, 1.000)	0.933 (0.846, 1.000)
AUC-SD	0.859 (0.821, 0.896)	0.952 (0.919, 0.985)	0.997 (0.991, 1.000)	0.937 (0.862, 1.000)
AUC-PD	0.934 (0.913, 0.956)	0.981 (0.963, 0.999)	1.000 (1.000, 1.000)	0.996 (0.985, 1.000)
AUC-micro	0.903 (0.872, 0.928)	0.964 (0.939, 0.985)	0.999 (0.995, 1.000)	0.956 (0.886, 0.996)
AUC-macro	0.915 (0.888, 0.937)	0.970 (0.950, 0.988)	0.999 (0.996, 1.000)	0.955 (0.895, 0.997)
F1-PR	0.854 (0.820, 0.883)	0.900 (0.835, 0.941)	0.889 (0.780, 0.947)	0.545 (0.382, 0.700)
F1-SD	0.826 (0.789, 0.858)	0.891 (0.823, 0.934)	0.939 (0.844, 0.978)	0.800 (0.639, 0.901)
F1-PD	0.833 (0.796, 0.864)	0.824 (0.747, 0.881)	0.978 (0.899, 0.995)	0.909 (0.767, 0.968)
Accuracy	0.834 (0.798, 0.865)	0.879 (0.810, 0.925)	0.946 (0.854, 0.982)	0.794 (0.632, 0.897)
New Lesion Assessment (benchmark)				
Precision	0.730 (0.688, 0.768)	0.852 (0.779, 0.904)	1.000 (0.936, 1.000)	Not applicable
Recall	0.844 (0.808, 0.874)	0.920 (0.859, 0.956)	1.000 (0.936, 1.000)	
F1-score	0.711 (0.668, 0.750)	0.885 (0.816, 0.930)	1.000 (0.936, 1.000)	
Accuracy	0.904 (0.875, 0.928)	0.952 (0.898, 0.978)	1.000 (0.936, 1.000)	
Overall Treatment Response Outcomes (benchmark)				
F1-PR	0.838 (0.802, 0.869)	0.889 (0.821, 0.933)	0.889 (0.780, 0.947)	0.545 (0.382, 0.700)
F1-SD	0.817 (0.780, 0.850)	0.885 (0.817, 0.930)	0.939 (0.844, 0.978)	0.800 (0.639, 0.901)
F1-PD	0.857 (0.823, 0.886)	0.914 (0.851, 0.951)	0.978 (0.899, 0.995)	0.909 (0.767, 0.968)
Accuracy	0.837 (0.800, 0.867)	0.895 (0.829, 0.938)	0.946 (0.854, 0.982)	0.794 (0.632, 0.897)
<i>CNN-based RECORD</i>				
SOV-based Tumor Burden Change (+3.07%, $P < 0.001$)				
AUC-PR	0.976 (0.962, 0.989)	0.997 (0.993, 1.000)	1.000 (1.000, 1.000)	0.975 (0.919, 1.000)
AUC-SD	0.904 (0.875, 0.933)	0.977 (0.957, 0.997)	1.000 (1.000, 1.000)	0.975 (0.936, 1.000)
AUC-PD	0.960 (0.944, 0.976)	0.985 (0.969, 1.000)	1.000 (1.000, 1.000)	1.000 (1.000, 1.000)
AUC-micro	0.937 (0.917, 0.955)	0.983 (0.966, 0.995)	1.000 (1.000, 1.000)	0.983 (0.945, 1.000)
AUC-macro	0.946 (0.929, 0.962)	0.986 (0.972, 0.996)	1.000 (1.000, 1.000)	0.983 (0.944, 1.000)
F1-PR	0.876 (0.843, 0.903)	0.945 (0.890, 0.974)	1.000 (0.936, 1.000)	0.600 (0.433, 0.746)
F1-SD	0.857 (0.823, 0.886)	0.923 (0.862, 0.958)	0.962 (0.875, 0.989)	0.865 (0.713, 0.943)
F1-PD	0.866 (0.832, 0.894)	0.840 (0.765, 0.894)	0.955 (0.865, 0.986)	0.952 (0.825, 0.988)
Accuracy	0.864 (0.830, 0.892)	0.911 (0.848, 0.950)	0.964 (0.879, 0.990)	0.853 (0.699, 0.936)
New Lesion Assessment (+3.23%, $P < 0.001$)				
Precision	0.880 (0.848, 0.907)	0.920 (0.859, 0.956)	1.000 (0.936, 1.000)	Not applicable
Recall	0.844 (0.808, 0.874)	0.920 (0.859, 0.956)	1.000 (0.936, 1.000)	
F1-score	0.741 (0.700, 0.779)	0.920 (0.859, 0.956)	1.000 (0.936, 1.000)	
Accuracy	0.945 (0.920, 0.962)	0.968 (0.920, 0.987)	1.000 (0.936, 1.000)	
Overall Treatment Response Outcomes (+3.50%, $P < 0.001$)				
F1-PR	0.867 (0.833, 0.895)	0.960 (0.910, 0.983)	1.000 (0.936, 1.000)	0.600 (0.433, 0.746)
F1-SD	0.859 (0.824, 0.887)	0.923 (0.862, 0.958)	0.962 (0.875, 0.989)	0.865 (0.713, 0.943)
F1-PD	0.892 (0.861, 0.917)	0.914 (0.851, 0.951)	0.955 (0.865, 0.986)	0.952 (0.825, 0.988)
Accuracy	0.873 (0.839, 0.900)	0.927 (0.868, 0.961)	0.964 (0.879, 0.990)	0.853 (0.699, 0.936)

95% CI is calculated.

Improvements of CNN-based RECORD on nnU-Net segmentation models were calculated on overall accuracy in all cohorts. P -value was calculated using McNemar's mid-p test.

PR response, SD stable disease, PD progressive disease.

Methods

Cohorts construction and ethics

Cohort A has the official title of "RATIONALE-208: A Phase 2, Open-label, Multicenter Study to Investigate the Efficacy, Safety, and Pharmacokinetics of the Anti-PD-1 Monoclonal Antibody BGB-A317 in Patients with Previously Treated Hepatocellular Unresectable Carcinoma" (NCT03419897).

Cohort B has the official title of "A Single-Arm, Multicenter Phase 2 Study of BGB-A317 in Patients with Previously Treated PD-L1+ Locally Advanced or Metastatic Urothelial Bladder Cancer" (NCT04004221). Detailed information is available at <https://clinicaltrials.gov/>. Cohort C was recruited from XinHua Hospital Affiliated to Shanghai Jiao Tong University. The study data were fully de-identified, and the patient's confidential information was

Table 3 | Comparisons of treatment response outcomes on validation sets (Swin-Unetr vs VIT-based RECORD)

Metrics	Cohort A1	Cohort A2	Cohort B	Cohort C
<i>Swin-Unetr</i>				
SOV-based Tumor Burden Change (benchmark)				
AUC-PR	0.944 (0.920, 0.968)	0.991 (0.981, 1.000)	1.000 (1.000, 1.000)	0.917 (0.807, 1.000)
AUC-SD	0.826 (0.786, 0.866)	0.954 (0.920, 0.989)	1.000 (1.000, 1.000)	0.961 (0.909, 1.000)
AUC-PD	0.918 (0.891, 0.945)	0.985 (0.969, 1.000)	1.000 (1.000, 1.000)	0.992 (0.973, 1.000)
AUC-micro	0.877 (0.849, 0.905)	0.969 (0.943, 0.990)	1.000 (1.000, 1.000)	0.966 (0.908, 0.997)
AUC-macro	0.892 (0.867, 0.917)	0.977 (0.957, 0.992)	1.000 (1.000, 1.000)	0.957 (0.896, 0.995)
F1-PR	0.786 (0.746, 0.820)	0.844 (0.770, 0.897)	1.000 (0.936, 1.000)	0.600 (0.433, 0.746)
F1-SD	0.769 (0.729, 0.805)	0.875 (0.805, 0.922)	1.000 (0.936, 1.000)	0.882 (0.734, 0.953)
F1-PD	0.816 (0.779, 0.849)	0.893 (0.826, 0.936)	1.000 (0.936, 1.000)	0.917 (0.777, 0.972)
Accuracy	0.788 (0.749, 0.822)	0.871 (0.801, 0.919)	1.000 (0.936, 1.000)	0.853 (0.699, 0.936)
New Lesion Assessment (benchmark)				
Precision	0.614 (0.569, 0.657)	0.786 (0.705, 0.849)	0.933 (0.836, 0.975)	Not applicable
Recall	0.844 (0.808, 0.874)	0.880 (0.811, 0.926)	1.000 (0.936, 1.000)	
F1-score	0.711 (0.668, 0.750)	0.830 (0.754, 0.886)	0.966 (0.881, 0.991)	
Accuracy	0.860 (0.826, 0.888)	0.927 (0.868, 0.961)	0.982 (0.906, 0.997)	
Overall Treatment Response Outcomes (benchmark)				
F1-PR	0.752 (0.711, 0.789)	0.857 (0.785, 0.908)	0.875 (0.764, 0.938)	0.600 (0.433, 0.746)
F1-SD	0.738 (0.696, 0.775)	0.865 (0.794, 0.914)	0.980 (0.903, 0.996)	0.882 (0.734, 0.953)
F1-PD	0.820 (0.782, 0.852)	0.914 (0.851, 0.951)	0.978 (0.899, 0.995)	0.917 (0.777, 0.972)
Accuracy	0.773 (0.733, 0.808)	0.879 (0.810, 0.925)	0.964 (0.879, 0.990)	0.853 (0.699, 0.936)
<i>VIT-based RECORD</i>				
SOV-based Tumor Burden Change (+2.63%, $P = 0.003$)				
AUC-PR	0.969 (0.951, 0.987)	0.987 (0.972, 1.000)	1.000 (1.000, 1.000)	0.967 (0.908, 1.000)
AUC-SD	0.854 (0.817, 0.892)	0.959 (0.927, 0.991)	1.000 (1.000, 1.000)	0.979 (0.943, 1.000)
AUC-PD	0.926 (0.902, 0.950)	0.987 (0.972, 1.000)	1.000 (1.000, 1.000)	1.000 (1.000, 1.000)
AUC-micro	0.901 (0.874, 0.927)	0.971 (0.946, 0.990)	1.000 (1.000, 1.000)	0.984 (0.946, 1.000)
AUC-macro	0.915 (0.892, 0.937)	0.978 (0.958, 0.992)	1.000 (1.000, 1.000)	0.982 (0.942, 1.000)
F1-PR	0.840 (0.804, 0.870)	0.877 (0.808, 0.924)	1.000 (0.936, 1.000)	0.667 (0.499, 0.801)
F1-SD	0.808 (0.770, 0.841)	0.905 (0.841, 0.945)	0.980 (0.903, 0.996)	0.895 (0.749, 0.960)
F1-PD	0.822 (0.785, 0.854)	0.889 (0.821, 0.933)	0.978 (0.899, 0.995)	0.952 (0.825, 0.988)
Accuracy	0.820 (0.782, 0.852)	0.895 (0.829, 0.938)	0.982 (0.906, 0.997)	0.882 (0.734, 0.953)
New Lesion Assessment (+1.54%, $P < 0.001$)				
Precision	0.667 (0.623, 0.708)	0.786 (0.705, 0.849)	0.933 (0.836, 0.975)	Not applicable
Recall	0.833 (0.797, 0.864)	0.880 (0.811, 0.926)	1.000 (0.936, 1.000)	
F1-score	0.741 (0.700, 0.779)	0.830 (0.754, 0.886)	0.966 (0.881, 0.991)	
Accuracy	0.881 (0.849, 0.907)	0.927 (0.868, 0.961)	0.982 (0.906, 0.997)	
Overall Treatment Response Outcomes (+3.07%, $P < 0.001$)				
F1-PR	0.820 (0.783, 0.852)	0.889 (0.821, 0.933)	1.000 (0.936, 1.000)	0.667 (0.499, 0.801)
F1-SD	0.784 (0.745, 0.819)	0.877 (0.808, 0.924)	0.980 (0.903, 0.996)	0.895 (0.749, 0.960)
F1-PD	0.834 (0.798, 0.865)	0.900 (0.835, 0.941)	0.978 (0.899, 0.995)	0.952 (0.825, 0.988)
Accuracy	0.811 (0.773, 0.844)	0.887 (0.819, 0.932)	0.982 (0.906, 0.997)	0.882 (0.734, 0.953)

95% CI is calculated.

Improvements of VIT-based RECORD on Swin-Unetr segmentation models were calculated on overall accuracy in all cohorts. P -value was calculated using McNemar's mid-p test.

PR response, SD stable disease, PD progressive disease.

removed. Therefore, the study was considered exempt from informed consent of the study participants. Cohort C consists of liver cancer patients treated with lenvatinib, bevacizumab, camrelizumab, or tislelizumab from 2017 to 2022. Data in Cohort A and B passed ethics review (application number: I2021173I) and were approved by the Human Genetic Resource Administration of China (approval number: [2021] GH5565). Data in Cohort C was approved by the Ethics Committee of Xinhua Hospital

(Approval No. XHEC-D-2023-081). The study was conducted in accordance with the Declaration of Helsinki.

Image processing

All patients underwent multi-slice helical CT scans to get portal venous phase scan using various scanners. For Cohort A and B, detailed scanner types include Philips Brilliance ICT 256/64, GE Lightspeed VCT, Discovery

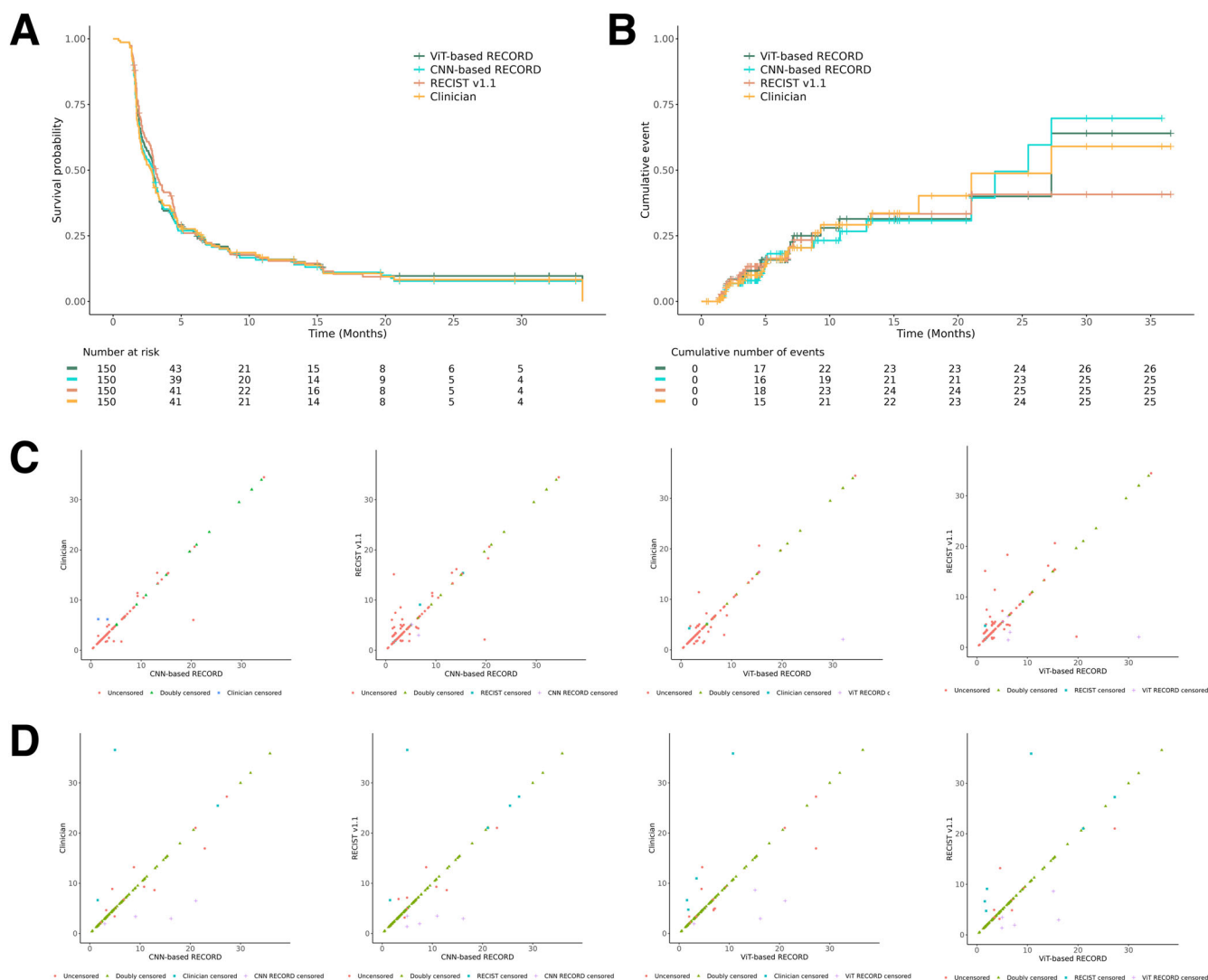


Fig. 6 | Comparisons of RECIST v1.1, clinician-annotated, and RECORD-predicted response and progression-free survival (PFS) time. A Kaplan–Meier curves of PFS. **B** Kaplan–Meier curves of time to response. **C** Pairwise comparisons of PFS time. **D** Pairwise comparison of time to response.

CT750 HD, Ingenuity CT, Revolution HD, Revolution EVO/GSI, Optima CT 540/660/680, Aquilion Prime, Aquilion One, BrightSpeed, uCT 530/550/760/780, and IQon Spectral CT. The liver was imaged using tri-phasic scans (i.e., late arterial phase, portal venous phase, and delayed/equilibrium phase). The methodology was consistent across visits for a subject (i.e., phases acquired, timing for each phase, etc). Scanning details were set by each separate site. For cohort C, CT scans were performed with a 64-channel MDCT (Somatom Definition, Siemens, Forchheim, Germany). Each patient fasted for more than four hours before undergoing CT scanning. The CT parameters were as follows: detector collimation (1 mm), pitch (0.9), gantry rotation (0.5), tube voltage (120 kV), tube current (240 mA), matrix (512 × 512), slice thickness (5 mm), and reconstruction interval (1 mm). The portal venous phase scanning was obtained using a fixed 65 s equilibrium following intravenous injection of 100 mL of ionic contrast material at a rate of 3 mL/s using an automatic injector and scan ranging from diaphragm to iliac crest.

We processed raw DICOM series to NIFTI format, normalized to [−200, 200] and resampled using SimpleITK⁴⁰ in Python. The original resolutions varied and were resampled to 0.8 mm × 0.8 mm × 5 mm. Data preprocessing is shown in Supplementary Fig. 14.

The ground truth 3D segmentation of each CT scan is blindly performed by two oncologists (both had 3 years of experience) using the ITK-

SNAP software (version 3.8.2). Only liver lesions were included and annotated. New lesions on follow-up scans were extra labeled (with tumor masks of longitudinal lesions being marked as 1, and new lesions marked as 2). During the annotation process, the two oncologists considered the patient's baseline condition and treatment history. All cases will be handed over to a third oncologist with 11 years of experience. The third oncologist primarily selects one of the results from the two oncologists. If necessary, this oncologist will modify (re-annotate) the mask based on one of the segmentation results. The treatment outcome assessment results are derived from the annotated tumor masks (see 'Definition of objective response outcomes'). All treatment response outcomes will be reviewed by a fourth medical imaging processing expert based on 3D CT scan images of the liver region.

Intra-rater variability was assessed. The DICE coefficient between the initial segmentations of the two assessors was 0.702 [SD 0.233]. The Fleiss Intraclass Correlation Coefficient⁴¹ (ICC [2,1]) for tumor volumes initially segmented by the two assessors was 0.924 (95% CI: 0.871, 0.964). For the treatment response outcome between the two initial assessors, ICC was 0.708 (95% CI: 0.662, 0.745), the ordinal Krippendorff's alpha⁴² was 0.785 (95% CI: 0.750, 0.820). After the third rater selected the more appropriate segmentation results, the treatment outcome results were calculated. The fourth rater had checked the treatment outcome results and reported no

Table 4 | Correlations among time to response and PFS predicted by different measures

	C-Index	Spearman-HighestRank	Spearman-restricted
PFS Time			
VIT-RECORD vs Clinician	0.928 (0.894, 0.962)	0.906 (0.846, 0.966)	0.828 (0.844, 0.968)
VIT-RECORD vs RECIST	0.874 (0.828, 0.921)	0.800 (0.711, 0.889)	0.800 (0.705, 0.895)
CNN-RECORD vs Clinician	0.958 (0.929, 0.987)	0.942 (0.887, 0.997)	0.942 (0.887, 0.997)
CNN-RECORD vs RECIST	0.886 (0.843, 0.929)	0.828 (0.749, 0.908)	0.828 (0.730, 0.927)
Clinician vs RECIST	0.867 (0.815, 0.918)	0.787 (0.689, 0.884)	0.787 (0.671, 0.902)
Response time			
VIT-RECORD vs Clinician	0.945 (0.894, 0.995)	0.763 (0.416, 1.000)	0.766 (0.451, 1.000)
VIT-RECORD vs RECIST	0.933 (0.871, 0.994)	0.664 (0.462, 0.867)	0.844 (0.579, 1.000)
CNN-RECORD vs Clinician	0.954 (0.908, 1.000)	0.765 (0.419, 1.000)	0.942 (0.644, 1.000)
CNN-RECORD vs RECIST	0.944 (0.890, 0.998)	0.797 (0.595, 0.998)	0.812 (0.531, 1.000)
Clinician vs RECIST	0.954 (0.911, 0.996)	0.726 (0.511, 0.942)	0.665 (0.262, 1.000)

95% CI is calculated by bootstrapping. All values in the table are significant ($P < 0.001$).

adjustment. The agreements between the final treatment assessment results and the initial results from the first two raters were also calculated. For the first rater, the ICC was 0.826 (95% CI: 0.794, 0.858), and Krippendorff's alpha was 0.880 (95% CI: 0.853, 0.904). For the second rater, the ICC was 0.909 (95% CI: 0.885, 0.931), and Krippendorff's alpha was 0.939 (95% CI: 0.921, 0.955).

Training data split

For model training and internal validation, we used five-fold cross-validation. We randomly divided cohort A1 into five folds by patients. There are four folds of 22 patients with 116, 120, 121, and 109 CT scans, respectively, and one-fold with 116 CT scans of 23 patients.

Definition of objective response outcomes

The treatment outcome can be assessed by comparing each follow-up scan to the baseline scan. The outcome evaluation considers both the change in tumor burden and the appearance of any new lesions. The sum of the volume (SOV) is used to reflect the change in tumor burden. A mapping function is designated to transform the SOV of liver lesions in both the baseline and follow-up into the probability of three response outcome classes.

The designed mapping function can be written as a combination of relative SOV change and absolute SOV change. The ground truth of SOV-based tumor status classification was generated by $\max(P_{PR}, P_{SD}, P_{PD})^T$. We visualized the mapping function by plotting a series of baseline and follow-up SOVs to show the change in three-class probability

(Supplementary Fig. 15).

$$\begin{pmatrix} P_{PR} \\ P_{SD} \\ P_{PD} \end{pmatrix} = w \cdot \begin{pmatrix} \sigma\left(C_1 \cdot \left(r_{PR} - \frac{V_2}{V_1}\right)\right) \\ \sigma\left(C_1 \cdot \left(r_{PD} - \frac{V_2}{V_1}\right)\right) - \sigma\left(C_1 \cdot \left(r_{PR} - \frac{V_2}{V_1}\right)\right) \\ 1 - \sigma\left(C_1 \cdot \left(r_{PD} - \frac{V_2}{V_1}\right)\right) \end{pmatrix} + (1 - w) \cdot (0, 1, 0)^T \quad (1)$$

where,

- r_{PR} , r_{PD} is extrapolated from 1D RECIST to 3D according to our dataset, where we set $r_{PR} = 0.65$, $r_{PD} = 1.44$. The calculation of these two critical values is described in “Validations on the designed mapping function”.
- C_i is a constant, where we set $C_1 = 15$, $C_2 = 5$, $C_3 = 0.1$;
- σ is the Sigmoid function, which maps $[-\infty, +\infty]$ to $[0, 1]$;
- V_1 is the baseline SOV of all liver lesions, V_2 is the follow-up SOV of all liver lesions;
- w is the trade-off between relative and absolute volume change, $w = \sigma\left(C_2 \cdot \log\left(\frac{|V_2 - V_1|}{V}\right)\right) + C_3 \cdot \max\left(\frac{V_2}{V}, \frac{V_1}{V}\right)$, which works only if both the baseline and follow-up lesions are small. V is the absolute volume required for progression, where we set $V = 400$. The two items are in line with the RECIST guideline, in which a change in diameter of less than 5 mm would be deemed stable disease;
- When the SOV is not particularly small (both SOV of baseline and follow are smaller than $\sim 1 \text{ cm}^3$), the contribution of the second term can be neglected. The argmax of the three-class probability vector should be equivalent to: a 44% increase indicates progression, and a 35% decrease indicates response.

New lesions were evaluated by radiologists. Ground truth for the treatment response evaluation integrated the result of both tumor burden classification and new lesion identification, with the rule shown in Supplementary Table 1.

Validations on the designed mapping function

The mapping function we designed referenced the existing 3D volumetric lesion measurement methods. In the existing literature^{43,44}, there are two different thresholds for progression and response evaluation—one assuming the lesions are spherical (in which case, progression is defined as a 73% or more increase in volume, and response is defined as a 65% or more decrease in volume), and the other assuming the lesions are ellipsoidal (in which case, progression is defined as a 20% or more increase in volume, and response is defined as a 35% or more decrease in volume). However, in actual data, lesions may not ideally be spherical or ellipsoidal, so using any one of the two thresholds could introduce potential biases. Consequently, we aimed to establish the critical volume change thresholds for progression and response by calculating the average volume ratio change for a single lesion when its diameter increased by 20% or decreased by 30% in our dataset. Cohort A was conducted with manual diameter measurements and was therefore used for mapping function modeling and validation. Specifically, we selected the single lesion in Cohort A1 that had a diameter reduction of 25–35%, and calculated the average volume change ratio of 0.6508 ± 0.1051 . Similarly, lesions with a diameter increase of 15–25% yielded an average volume change ratio of 1.4403 ± 0.2282 . Validation results in Cohort A2 showed that the average volume change ratio was 0.6398 ± 0.0531 , and 1.4639 ± 0.1379 for response and progression, respectively. The similar volume change rates of Cohorts A2 with A1 indicate the potential generalizability of the critical value we have set.

With selected thresholds for progression and response, our mapping function has been further designed with smoothing for the three-class classification problem. In the scenario where the tumor volume

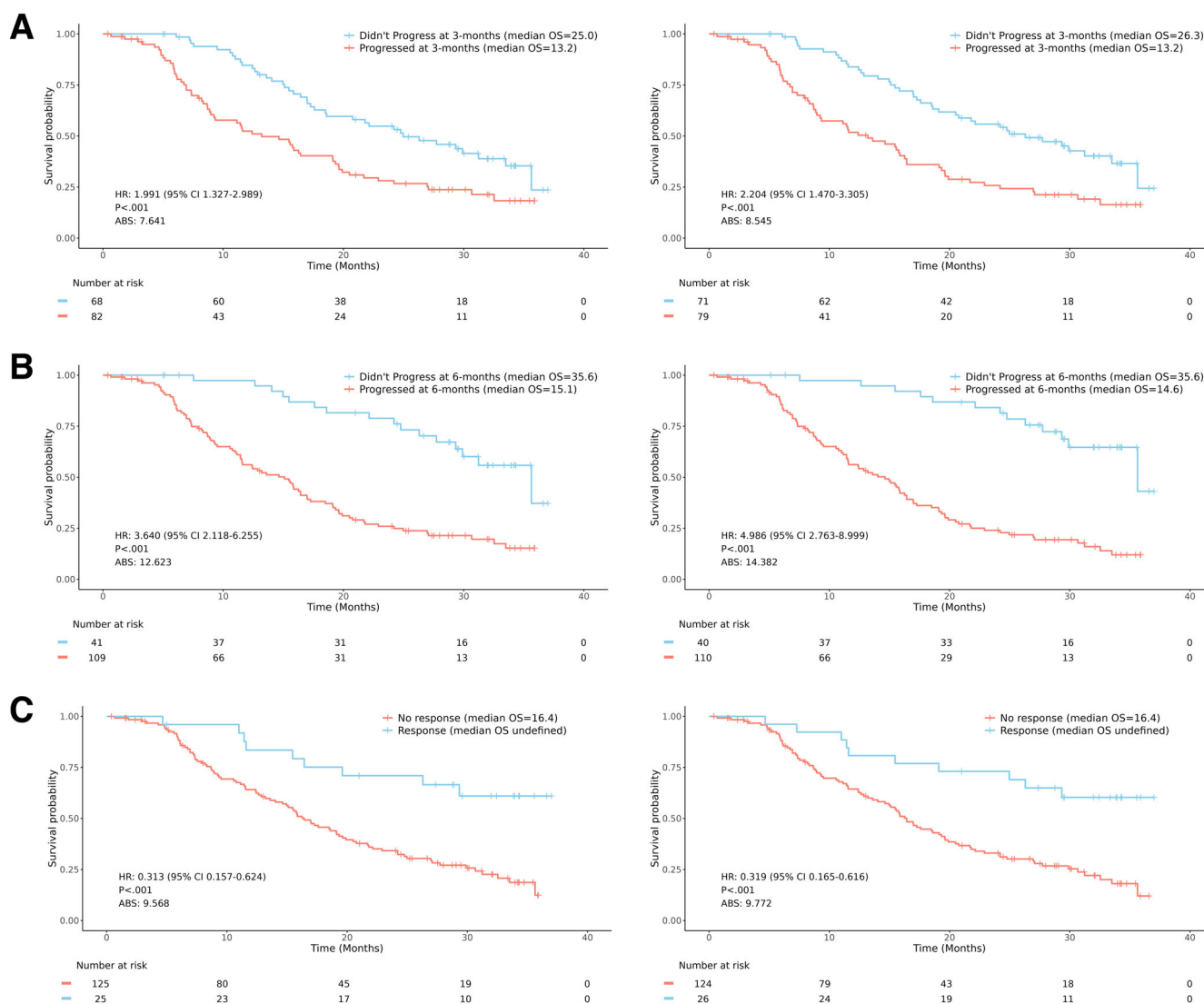


Fig. 7 | Kaplan-Meier curves for OS stratified by the treatment response status. Left, CNN-based RECORD; Right, ViT-based RECORD. **A** Stratification by progression at a landmark time of 3 months. **B** Stratification by progression at

landmarking time of 6 months. **C** Stratification by response during the treatment. ABS area between survival curves, HR hazard ratio, OS overall survival.

increases by 44%, the probabilities of stability and progression are exactly equal. When the tumor volume increases by more than 44%, the treatment evaluation result will be disease progression. The same logic applies for the response.

We compared automatically calculated response outcomes from human-annotated tumor masks to fully human-assessed RECIST v1.1 on Cohort A. The two measures may have potential differences because the former considered all tumors while the latter considered the two largest tumors. We have calculated the consensus of the two measures: The automatically calculated response achieved 81.2% consensus with the RECIST v1.1 for response outcomes (Supplementary Fig. 16).

RECORD architecture

The tumor segmentation step adopts both CNN-based nnU-Net and ViT-based Swin-Unetr. For deformable registration, we re-implemented Advanced Normalization Tools (ANTs)⁴⁵ in PyTorch to make deformable transformations differentiable. Both forward and inverse displacement fields for follow-up images were generated by ANTs. Since we are to evaluate tumor burden based on volume, the inverse consistency of the registration process on volume was verified (Supplementary Fig. 17).

The workflow for difference map generation is described in Supplementary Algorithm 2. After registration of the predicted masks, we subtracted the baseline mask from the follow-up mask, and obtained differences on single lesion level by 3D connected component analysis. Then, we marked any extremely large differences as potential false positives and checked whether they represented actual disease progression or model mispredictions, using the correlated information from other serial CT scans. In addition, a liver mask predicted by the union of “livermask”⁴⁶ and pre-trained nnU-Net on abdominal organ segmentation task⁴⁷ was added to remove lesions located outside the liver, further guiding the optimization model to learn relevant features from the original predicted tumor mask. The following paragraph provides more implementation details.

The difference map generation involves three key steps: checking suspected abnormal large lesions (Supplementary Algorithm 2, line 11–14), checking abnormal growth trend (Supplementary Algorithm 2, line 15–24), and removing lesions outside the liver (Supplementary Algorithm 2, line 27). The first step of checking abnormally large lesions is a rough selection of lesions that have drastically changed between the baseline and follow-up CT scans. To distinguish whether this is natural disease progression or a misprediction by the segmentation models, the variation between the predicted second-order lesion masks is then calculated. Second-order lesion masks

refer to the differential of the differences between predicted tumor masks at two consecutive time points. Larger variation of a particular lesion is more likely to indicate a misprediction, as tumors have a low probability of shrinking or expanding significantly in a short period. A lesion with extremely large second-order variation would be removed on the difference map. Finally, the liver mask is used to remove any predicted lesions that are located outside the liver. The thresholds for these steps are set to be excessive, in order to retain as much original information as possible and only remove extreme cases of misleading segmentation predictions.

The SOV-based treatment response classification model was built on a multi-task learning framework that considered both the accuracy of lesion segmentation and progression classification (Fig. 3B, upper branch, blue dotted line, L_{dice} and $L_{progression}$). The whole procedure was described in Supplementary Algorithm 1. The model received inputs from the baseline and follow-up tumor masks predicted by the segmentation model (i.e., CNN-based nnU-Net or ViT-based Swin-Unetr) as well as the difference map generated by the predicted masks. Then, the model took the concatenation of the baseline mask, the generated difference map, and the registered follow-up mask as input, and fed it into an encoder-decoder architecture with two output heads - a segmentation head and a classification head. For the segmentation head, traditional DICE and cross-entropy loss was used to ensure accurate tumor segmentations. For the classification head, an ordinal regression approach was used to transform the SOV of baseline and follow-up tumor masks into a three-class probability vector. During the decision-making stage of the model, the argmax of the predicted probability vector was used to determine whether the treatment outcome predicted by the model was response, stability, or progression. For the encoder-decoder structure, we utilized the Squeeze and Excitation Unet (Se-Unet). Se-Unet incorporated squeeze and excitation (SE) blocks⁴⁸ into the 3D U-Net⁴⁹ architecture. To balance computational complexity and avoid overfitting, three encoder-decoder blocks and four bottleneck blocks were used.

For multi-task learning, the model was optimized with gradients for both accurate segmentation and classification. We designed the loss function to be a weighted sum of the segmentation loss (DICE loss and cross-entropy loss) and the progression loss (ordinal regression).

$$L = L_{seg} + w \cdot L_{progression}, w = 0.55 \quad (2)$$

$$L_{seg} = L_{seg_baseline} + L_{seg_followup} \quad (3)$$

$$L_{progression} = -y \cdot \ln \hat{y}, \hat{y} = \frac{e^{-x_i}}{\sum_{k=1}^3 e^{-x_k}} i = 1, 2, 3 \quad (4)$$

where w is the trade-off between two tasks. Based on our experiments, we recommend setting w between 0.4 and 0.7, and found that $w = 0.55$ achieve the best performance. L_{seg} represents the segmentation loss for both the baseline and follow-up images. $L_{progression}$ represents the progression loss, which depicts the classification accuracy of response evaluations. It is crucial to highlight that the ground truth label y is designed to be continuous in the model optimization, without employing argmax that is typically utilized in traditional classification problems. This is because there exists an order among response, stable, and progressive, thus regarding this as an ordinal regression problem, rather than a traditional classification problem, can give more tolerance to corner cases.

The SOV-based treatment response classification model was implemented using MONAI 0.9.1, a PyTorch-based repository. To handle the computation load of 3D image pairings, we adopted data parallel, used sliding window strategies to distribute ROI patches computation to four GPUs.

Explanations on the design of ordinal regression-based

loss $L_{progression}$

The loss function of response outcome classification is based on ordinal regression, which is modified from the traditional categorical classification to in line with the continuous disease progression process. Traditional cross-entropy loss treats the ground truth label as three one-hot classes, while our modified loss function converts the discrete three-class label into a probability vector. The nature of treatment outcome is a continuous quantification of change ratio of tumor burden, rather than a discrete value determined by several cut-offs. Specifically, thresholds between disease statuses are manually set to get the results from the baseline-follow-up SOV change ratio, with a given percentage of SOV decrease denotes PR (in our scenario is 35%), and another given percentage increase in SOV denotes PD (in our scenario is 44%). One-hot discrete labels can have information loss. For instance, clinicians would consider 43% and 45% increase in SOV have little difference in disease progression status. However, if we adopt categorical classification during the optimization, the former would be one-hot encoded with (0, 1, 0) (SD), while the latter would be (0, 0, 1) (PD), leading to totally different ground truth response outcomes. The simplest way to avoid information loss is to directly predict the continuous tumor burden change ratio. Whereas, it is also inappropriate to treat the problem as a traditional regression task that employs a standard mean squared error (MSE) loss, since the tumor burden change ratio exhibits boundary effects. For example, extreme change ratios such as 20 versus 30 times greater may both be indicative of disease progression, yet they can cause large loss during model optimization when using MSE as the loss function. Conversely, small change ratios such as 0.5 (shrinkage) and two times (expansion) can result in vastly different response evaluations, but their contribution to the loss would be smaller than that of the former scenario. To precisely sum up, we carefully designed a mapping between SOV change ratio and the three-class probability vector, fully utilizing label information while matching the practical response evaluation scenario.

Longitudinal graph matching in the new lesion assessment. Graph matching was used to model the new lesion identification process in the RECORD model. The generated difference map was added to the registered masks to remove possible erroneous predictions. Subsequently, 3D connected component analysis was employed to obtain the individual lesions. Then, a longitudinal lesion graph $G = (L_B, L_F, E)$ was constructed, where $L_B = \{l_1^B, l_2^B, \dots, l_n^B\}$ and $L_F = \{l_1^F, l_2^F, \dots, l_n^F\}$ represented the baseline (B) and follow-up (F) lesions, respectively; E represented the edge between the baseline and follow-up lesions. In the new lesion assessment, we need to avoid erroneous positive diagnoses caused by lesion splitting or merging, especially in cases where lesion splitting is present. We assume that split or merged lesions are relatively close located. Therefore, during the graph construction, the relative Euclidean distance $c_{ij} = \frac{d_{ij}}{r_i^B + r_j^F}$ between baseline and follow-up lesions was calculated, where d_{ij} represented the center-to-center distance between two lesions, and r_i^B, r_j^F represented the radii of the baseline and the follow-up lesions, respectively. When $c_{ij} < 1.0$, an edge $e_{ij} = (l_i^B, l_j^F)$ was added to E . If all l_j^F in L_F had edge, there should be no new lesion assessed by the graph matching, otherwise there exist new lesion(s). Schematic representations are provided in Supplementary Fig. 18.

Model configurations. All the deep learning models were trained from scratch using random initialization. The tumor segmentation models, nnU-Net and Swin-Unetr, were trained on an NVIDIA A100 (40GB). The patch size for nnU-Net was adaptively learned based on the dataset characteristics, and for Swin-Unetr the patch size was $96 \times 96 \times 96$. Both the nnU-Net and Swin-Unetr models were trained for 1000 epochs.

As for the RECORD model, $4 \times$ NVIDIA A100 (40GB) GPUs were used to optimize the model at 3D full resolution for CT inputs. The learning rate was adaptively tuned throughout the training process. The initial learning rate

was set to $1e-4$, and then a cosine annealing learning rate strategy (the CosineAnnealingLR method in PyTorch) was employed to allow the learning rate to change periodically. The RECORD model was trained for 200 epochs. Minimum loss on the validation set was used to select the model.

Model ensemble

The five trained models from the 5-fold cross-validation on Cohort A1 were ensembled, and then used for inference on Cohort A2, B, and C. For segmentation, we averaged the output probability maps from the five models. For treatment outcome classification, the ensembled predictions were generated by averaging the three-class probabilities predicted by the five trained models. The class with the highest averaged probability was served as the final treatment assessment result.

Statistical analyses

The treatment outcome results for Cohort A1 were obtained by merging all five-fold cross-validation samples predicted by five independently trained models. The ensemble predictions of the models trained in five-fold cross-validation on Cohort A1 were employed as results for the cohorts A2, B, and C. For the SOV-based response classification, the area under the receiver operating characteristic curve (AUROC) was calculated for each class as well as on a micro and macro level. The 95% confidence interval (CI) of the AUROC for each class was calculated using the DeLong method, while the 95% CI of the micro and macro AUROC was calculated using bootstrapping. In addition, the F1-score was calculated for each class using a one-vs-rest strategy. The overall accuracy (which equals to F1-micro) was also reported. For the new lesion assessments, precision, recall, F1-score, and accuracy were calculated. For the overall treatment outcome evaluations, the F1-PR, F1-SD, F1-PD were also calculated using a one-vs-rest strategy, and the overall accuracy was reported. Calculation of the 95% CI of the F1-scores used the Wilson method⁵⁰. Significance testing of the AUC curves and F1-scores used the DeLong test⁵¹ and the McNemar mid-p test⁵², respectively.

For the endpoints of PFS and response time, we compared the measurements from four different approaches: the segmentation model, the RECORD model, tumor mask annotations from clinicians, and the manually assessed RECIST v1.1 results. All four methods were able to provide evaluations of the patient's disease status (response, stabilization, or progression) at each follow-up time point. The PFS and time to response were then determined based on these treatment response assessment results. PFS was defined as the time from the initial baseline assessment (when a patient underwent a CT scan and started the first treatment) to either the first observed progression of disease on a CT scan or the time of death, whichever occurs first. Time to response was defined as the duration from the baseline evaluation to the first observed response state observed on a CT scan. To quantify the agreement between PFS and response time provided by different measures, the concordance index (c-index)⁵³ and Spearman's correlation extended for survival data⁵⁴ were employed to analyze correlations in bivariate right-censored data.

For the analyses of clinical endpoints (PFS and response time) as surrogates for OS, landmarking survival analyses⁵⁵ were conducted. For PFS, landmarks were chosen the progression status at 3 months and 6 months from entry. For time to response, a specific landmarking time was not selected due to the limited number of early responses, instead by the response status during the treatment. The disease status at specific landmarking time of each patient was served as a covariate for the Cox model for analysis. A larger hazard ratio (HR) and higher area between the survival curves (ABS) indicates the effectiveness of PFS/response time as predictor(s) for OS.

Cohort A (comprising the five-fold validation results from Cohort A1 and test results from Cohort A2) was analyzed for both clinical endpoint and OS stratification evaluations, given that it comprises an adequate and appropriate number of samples (150 patients) with for the purpose of conducting survival analyses. Furthermore, all patients in this cohort had primary hepatocellular carcinoma and received the same treatment, thereby ensuring the uniformity of the samples.

Calculations for AUROC, precision, recall, F1-score, and accuracy used sklearn 0.22.2 in Python 3.7.13. The agreement measures for bivariate right-censored data used survSpearman 1.0.1 in R 4.0.2. The landmarking analyses used survival 3.1.12 and survminer 0.4.9 in R 4.0.2.

Data availability

The datasets cannot be made publicly available due to general data protection regulations and institutional guidelines. Data can be retrieved upon reasonable request.

Code availability

The study source code can be found at <https://github.com/EstelleXIA/RECORD/>.

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Competing interests

Z.S.Y., Y.J.X., JieZ., JinZ., X.L.X., and G.K. are filing a patent for the deep learning model in this study. All other authors declare they have no competing interests.

Additional information

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